

QFT – Lecture 1

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Aktarım notu: Kaynağın 1. sayfası yazısız turkuaz bir kapak, 9. sayfası boş kareli bir sayfadır. Ders içeriği 2–8. sayfalardadır. Metin özgün İngilizcesiyle aktarılmıştır. Gri aktarım notları özgün ders metnine ait değildir; kırmızı açıklamalar kaynakta kırmızı yazılmış ekleri gösterir. Ofis numarasındaki soru işaretleri kaynakta da bulunmaktadır.

I. Motivation

GOAL is to compute decay rates [illegible/erased fragment] for inelastic collision.

Aktarım notu: “Decay rates” ile “for” arasında silinmiş veya çok silik bir parça vardır; güvenilir biçimde okunamadığı için tamamlanmamıştır.

inelastic collision $\implies$ for any final states.

Context in QM

- The Dirac equation (spin- $\frac{1}{2}$).
- The Klein–Gordon equation (spin-0).

Blackboard text

But they deal with a particle only (and its antiparticle).

SOLUTION QFT is a quantum relativistic theory of fields.

(Particles will be modes of these fields.)

Features

Relativity

$$v \leq c$$

Causality.

Quantum theory

$$\text{probability} \propto |\text{probability amplitude}|^2.$$

probability amplitude

$$= \sum_{\text{channels}} \text{probability amplitude for the channel.}$$

$$\star \quad \sum \text{probability} = 1.$$

II. First encounter with the S-matrix

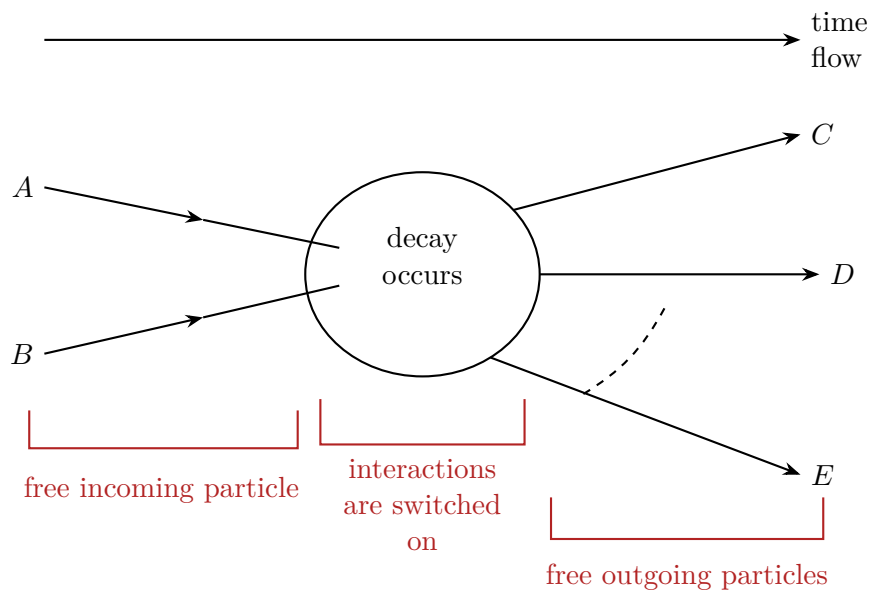
New goal: to compute probability amplitudes.

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Example: let's consider two-body decay

$$A + B \longrightarrow C + D + E + \dots$$

Aktarım notu: “Two-body decay” ifadesi ile yukarıdaki süreç kaynakta bu şekilde birlikte yazılmıştır; terim değiştirilmemiştir.

Picture**Quantum description**

- The incoming state: “ A and B ” system,

$$|\text{in}\rangle \quad (\text{normalized state}).$$

- “The decay occurs.”

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Quantum description – blackboard

- The incoming state: “ A & B ” system,

$$|\text{in}\rangle \quad (\text{normalized state}).$$

- “The decay occurs”:

$$\boxed{S|\text{in}\rangle}.$$

The state $S|\text{in}\rangle$ contains the information of all possible outgoing states.

Annotations pointing to S :

- Contains the interaction.
- Evolution operator which acts on $|\text{in}\rangle$.

Probability amplitude and probability

- The probability amplitude to get the “ C and D and E ” system (the $|\text{out}\rangle$ state) is

$$\langle \text{out} | S | \text{in} \rangle.$$

- Probability of this decay:

$$|\langle \text{out} | S | \text{in} \rangle|^2.$$

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Consequence

Assuming all the $|\text{out}\rangle$ states are linearly independent,

$$\left(\text{i.e. } \sum_{\text{out}} |\text{out}\rangle \langle \text{out}| = \mathbf{1} \right),$$

we have

$$\sum_{\text{out}} |\langle \text{out}|S|\text{in}\rangle|^2 = 1.$$

The left-hand side is

$$\begin{aligned} & \sum_{\text{out}} \underbrace{\langle \text{in}|S^\dagger|\text{out}\rangle}_{\langle \text{out}|S|\text{in}\rangle^*} \langle \text{out}|S|\text{in}\rangle. \\ & \quad \Downarrow \\ & \langle \text{in}|S^\dagger \left(\underbrace{\sum_{\text{out}} |\text{out}\rangle \langle \text{out}|}_{\mathbf{1}} \right) S|\text{in}\rangle. \end{aligned}$$

Hence

$$\langle \text{in}|S^\dagger S|\text{in}\rangle = 1.$$

So:

$$\boxed{S^\dagger S = \mathbf{1}} \quad \text{Unitarity.}$$

Aktarım notu: Kaynağın “linearly independent (i.e. . . .)” ifadesi aynen korunmuştur. Yazılan birim operatör bağıntısı, yalnızca doğrusal bağımsızlık değil, tam bir ortonormal durum kümesi varsayımıdır. Üniterlik sonucu, norm korunmasının tüm giriş durumları için geçerli olmasıyla anlaşılır.

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III. Some formal reminders

1. A few mathematical formulae

a) The Volterra equation – T-product

Definition: Volterra equation

$$\dot{F}(t) = f(t) \quad \Longrightarrow \quad F(t) = F(t_0) + \int_{t_0}^t f(\tau) \, d\tau.$$

Blackboard annotation: “Check this works.”

Example: the Schrödinger equation

$$i\hbar \dot{\psi}(t) = V(t)\psi(t) \quad \Longleftrightarrow \quad \underbrace{\dot{\psi}(t)}_{\dot{F}(t)} = \underbrace{\frac{1}{i\hbar} V(t)\psi(t)}_{f(t)}.$$

Let’s find another expression for the solution of the “S eq.”

$$\begin{aligned} \psi(t) &= \psi(t_0) + \int_{t_0}^t \frac{1}{i\hbar} V(t_1)\psi(t_1) \, dt_1 \\ &= \psi(t_0) + \int_{t_0}^t \frac{1}{i\hbar} V(t_1) \underbrace{\left\{ \psi(t_0) + \int_{t_0}^{t_1} \frac{1}{i\hbar} V(t_2)\psi(t_2) \, dt_2 \right\}}_{\psi(t_1)} \, dt_1. \end{aligned}$$

Aktarım notu: “S eq.” kısaltması burada Schrödinger denklemini belirtir. Alt satırdaki kırmızı ifade ve $\psi(t_1)$ alt açıklaması kaynakta vardır.

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We can repeat the same type of substitutions.

Blackboard: We can repeat the same type of substitutions “ad infinitum”:

$$\psi(t) = \psi(t_0) + \sum_{n=1}^{\infty} \psi(t_0) \left(\frac{1}{i\hbar} \right)^n \int_{(n)} V(t_1) V(t_2) \cdots V(t_n),$$

where

$$\int_{(n)} \equiv \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n.$$

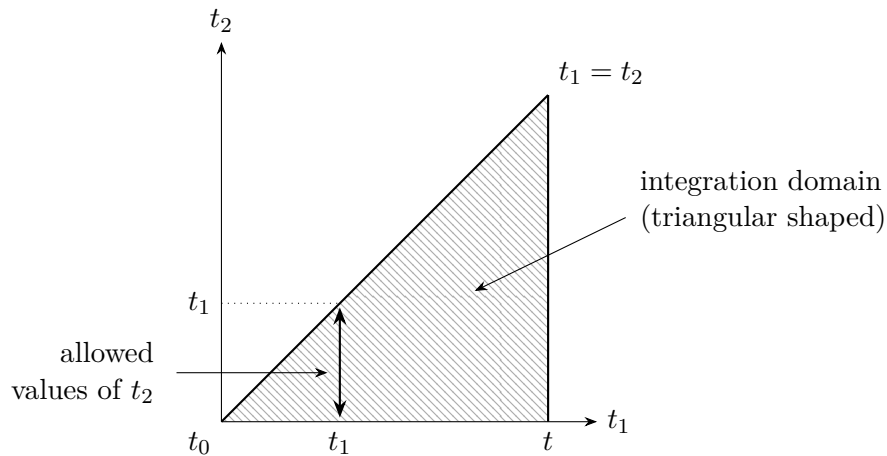
Aktarım notu: Bu ilk seri satırında $\psi(t_0)$ kaynakta operatör çarpımının solunda yazılmıştır ve burada aynen korunmuştur. Operatörlerin duruma etkisini gösteren, aynı sayfanın sonundaki ifadelerde $\psi(t_0)$ sağdadır.

Let’s consider the $n = 2$ term

$$J_2 = \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 V(t_1) V(t_2).$$

This implies that

$$t_0 \leq t_2 \leq t_1.$$



Integration domain of J_2 ; allowed values of t_2 .

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We want the boundaries of the integrals not to depend on the integration variables.

Answer: the Heaviside function,

$$\Theta(t) = \begin{cases} 1, & \text{when } t \geq 0, \\ 0, & \text{when } t < 0. \end{cases}$$

Then

$$J_2 = \int_{t_0}^t dt_1 \int_{t_0}^{\textcolor{red}{t}} dt_2 V(t_1) V(t_2) \Theta(t_1 - t_2).$$

Aktarım notu: İkinci integralin sabit üst sınırı t , kaynakta daire içine alınmıştır.

A few extra manipulations give

$$\begin{aligned} J_2 &= \frac{1}{2} J_2 + \frac{1}{2} J_2 \\ &= \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \frac{1}{2} V(t_1) V(t_2) \Theta(t_1 - t_2) + \text{“the same”}. \end{aligned}$$

Source annotation: $t_1 \rightarrow t_2$ and $t_2 \rightarrow t_1$.

$$J_2 = \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \frac{1}{2} \left(V(t_1) V(t_2) \Theta(t_1 - t_2) + V(t_2) V(t_1) \Theta(t_2 - t_1) \right).$$

We can now introduce the **T-product (time-ordered product)**.

For two operators:

$$\underbrace{\text{T}}_{\text{time}}[V(t_1) V(t_2)] = \begin{cases} V(t_1) V(t_2), & \text{when } t_1 \geq t_2, \\ V(t_2) V(t_1), & \text{when } t_2 > t_1. \end{cases}$$

However, using the Heaviside function:

$$\begin{aligned} \text{T}[V(t_1) V(t_2)] &= V(t_1) V(t_2) \Theta(t_1 - t_2) \\ &\quad + V(t_2) V(t_1) \Theta(t_2 - t_1). \end{aligned}$$

Hence:

$$\boxed{J_2 = \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \frac{1}{2} \text{T}[V(t_1) V(t_2)]}.$$

Aktarım notu: Kaynakta $\Theta(0) = 1$ yazılıdır. Heaviside ile verilen son ifade, eşit zamanlarda iki terimi de sayar; bu tek nokta konvansiyonu kaynakta ayrıca ele alınmamıştır. Formüller özgün halleriyle korunmuştur.

SOURCE PDF: PAGE 7, BOTTOM BLACKBOARD PHOTOGRAPH

For n in general

$$J_n = \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \cdots \int_{t_0}^t dt_n \frac{1}{n!} T[V(t_1)V(t_2) \cdots V(t_n)].$$

The solution to the “S eq.”

$$\begin{aligned} \psi(t) &= \psi(t_0) + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\frac{1}{i\hbar} \right)^n \underbrace{\int_{t_0}^t dt_1 \cdots \int_{t_0}^t dt_n T[V(t_1) \cdots V(t_n)]}_{\text{“}n \text{ dependency”}} \psi(t_0). \\ \Rightarrow \quad \psi(t) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{i\hbar} \right)^n \left\{ \int_{t_0}^t dt_1 \cdots \int_{t_0}^t dt_n T[V(t_1) \cdots V(t_n)] \right\} \psi(t_0). \end{aligned}$$

Source annotation: The factor $(1/(i\hbar))^n$ is inserted by red arrows into both series in the source.

Blackboard annotations:

$$n = 0 : \quad 1, \quad \sum_{n=0}^{\infty} \frac{1}{n!} x^n.$$

Aktarım notu: $n = 0$ terimindeki boş integral ve boş operatör çarpımı birim olarak yorumlanır; kaynakta bu “ $n = 0 : 1$ ” şeklinde belirtilmiştir.

SOURCE PDF: PAGE 8, TOP BLACKBOARD PHOTOGRAPH

Since this looks like the expansion of the exponential function, we define:

Time-ordered exponential

$$\begin{aligned} & \text{T exp} \left(\int_{t_0}^t f(\tau) d\tau \right) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \int_{t_0}^t dt_1 \cdots \int_{t_0}^t dt_n \text{T}[f(t_1) \cdots f(t_n)]. \end{aligned}$$

Hence, we can write the solution of the “S” equation as:

$$\begin{aligned} & i\hbar \dot{\psi}(t) = V(t)\psi(t) \\ & \iff \psi(t) = \text{T exp} \left(\frac{1}{i\hbar} \int_{t_0}^t V(\tau) d\tau \right) \psi(t_0). \end{aligned}$$

Remark: If V does not depend on time, then

$$\psi(t) = \underbrace{\exp \left(\frac{1}{i\hbar} (t - t_0) V \right)}_{\text{evolution operator in QM}} \psi(t_0).$$

b) Green functions

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A method used to solve differential equations.

We want to solve

$$Ly(x) = f(x),$$

where L is a differential operator

$$\left(\text{e.g. } L = \frac{d^2}{dx^2} \right).$$

If we can solve

$$LG(x) = \underbrace{\delta(x)}_{\text{Dirac "function"}}, \quad (*)$$

then

$$y(x) = \int_{-\infty}^{+\infty} G(x-x')f(x') dx' = \underbrace{(G * f)(x)}_{\text{convolution product}}.$$

The problem is to solve (*) (method by Fourier transform):

$$\text{FT}[\delta(x)] = 1.$$

G : Green function.

SOURCE PDF: PAGE 8, THIRD PHOTOGRAPH, RIGHT SIDE, AND BOTTOM PHOTOGRAPH

2. Canonical transformations in QM

a) Definition

A canonical transformation does not change the physical content of the theory and keeps the structure of the mathematical expression intact. It simplifies the calculation.

In QM, the canonical transformations will be unitary operators U .

States

$$|\psi\rangle \xrightarrow{U} |\psi'\rangle = |U\psi\rangle = U|\psi\rangle.$$

Remark:

$$\begin{aligned}\langle\chi'|\psi'\rangle &= \langle\chi|\underbrace{U^\dagger U}_1|\psi\rangle \\ &= \langle\chi|\psi\rangle.\end{aligned}$$

Side annotation:

$$\langle\chi'| \longleftarrow |\chi'\rangle^\dagger.$$

Observable (operator)

$$A \xrightarrow{U} A' = UAU^\dagger.$$

b) Reminder: Schrödinger picture

State

$|\psi(t)\rangle$

$$\begin{cases} i\hbar \frac{\partial |\psi\rangle}{\partial t} = H|\psi\rangle, \\ |\psi(t_0)\rangle = |\psi_0\rangle. \end{cases}$$

Aktarım notu: Ders içeriği kaynakta bu başlangıç koşuluyla sona erer. Sonraki 9. sayfa boştur.